The 7th Annual Parallel Applications Workshop, Alternatives to MPI+X

Intel® SHMEM: GPU-initiated OpenSHMEM using SYCL

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Motivation

- Distributed multi-GPU systems are growing rapidly
- 9 of the top 10 TOP500 systems include GPU accelerators
  - #1 – Frontier has 4 GPUs per node
  - #2 – Aurora has 12 Ponte Vecchio (PVC) GPU tiles on each node connected with Intel Xe Link Fabric and with dual Sapphire Rapids (SPR) CPUs
  - #3 – Eagle has 8 GPUs per node
- Current OpenSHMEM standard does not yet enable GPUs
- Vendor specific solutions exist that support specific hardware and programming model

https://singularityhub.com/2022/05/30/age-of-exascale-wickedly-fast-frontier-supercomputer-ushers-in-the-next-era-of-computing/

https://www.datacenterdynamics.com/en/news/doe-finally-starts-installing-intels-oft-delayed-aurora-supercomputer-now-expected-to-hit-2-exaflops/

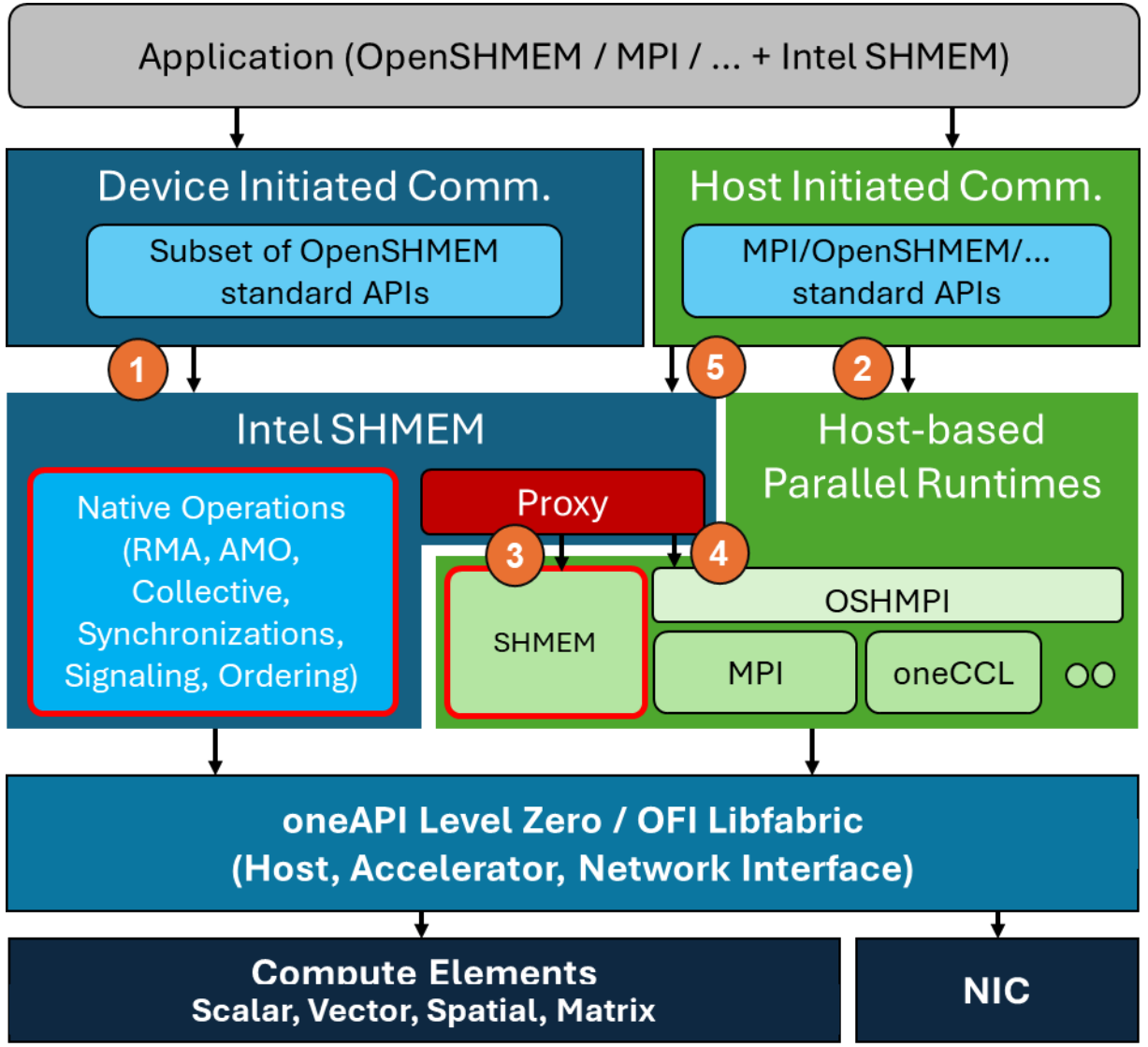
https://www.datacenterdynamics.com/en/news/microsoft-reveals-immense-cloud-supercomputer-just-openai/

Contribution

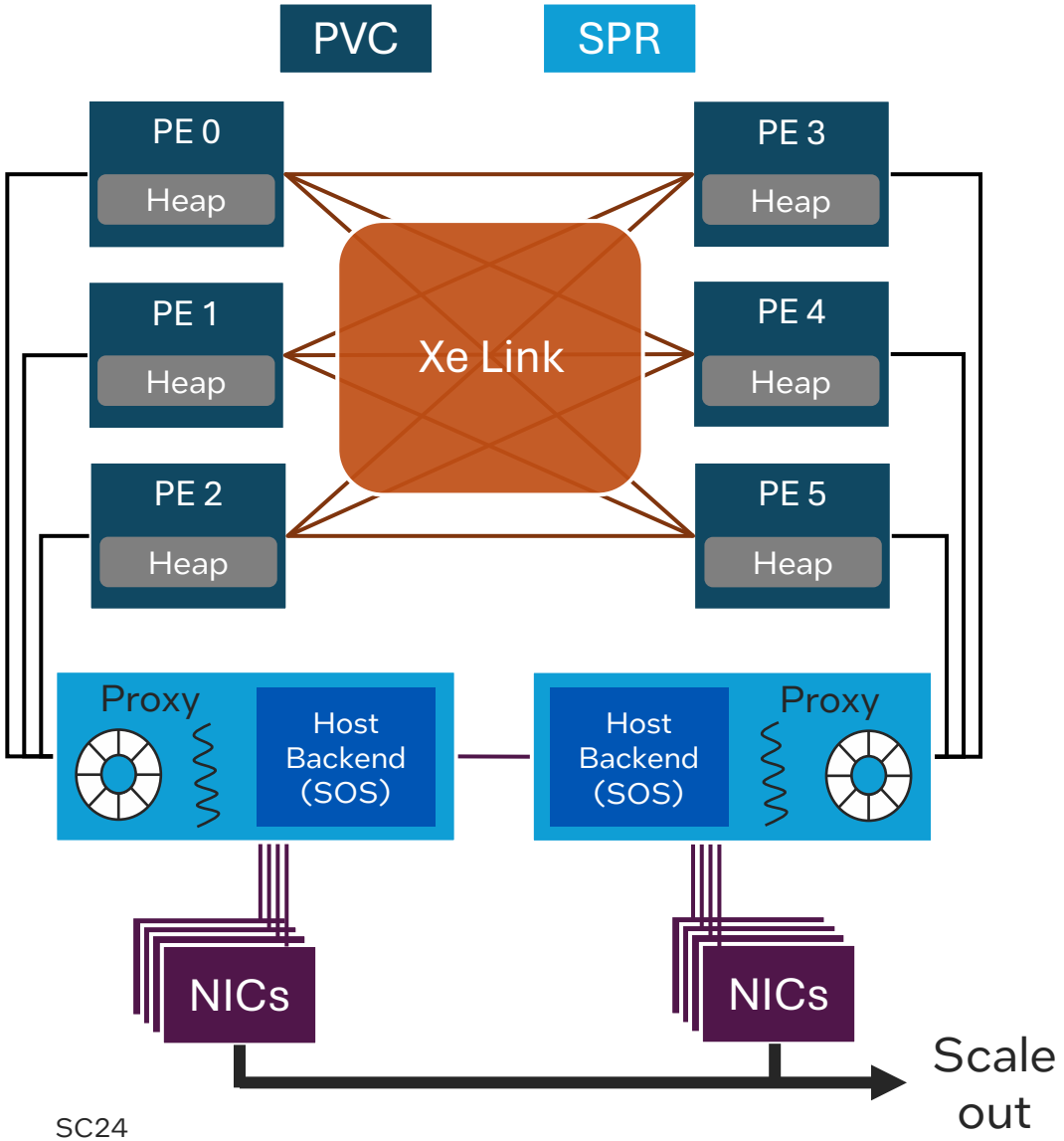
- Intel® SHMEM
  - A library for one-sided and collective communication
  - Implements the OpenSHMEM API, with extensions for GPU using SYCL
- Operations Supported
  - Remote memory access (RMA), atomic memory operations, collective operations, synchronization of multiple processing elements or PEs (i.e., MPI ranks), signaling operations – aligned with OpenSHMEM 1.5
  - API available on host and inside SYCL kernels (device initiated)
- Benefits
  - Simplified programming model for GPU-initiated operations
  - High-performance communications for distributed GPU applications
  - Can leverage high-performance host runtimes

Architecture Overview

1. Device-initiated OpenSHMEM operations
   - Subset of the standard APIs
   - Extensions to leverage GPU threads
   - Intra-node operations leverage L0 IPC for direct GPU to GPU transfer
2. All host back-end operations are supported as it is
3. Inter-node device-initiated operations utilize a host proxy and SHMEM runtime
4. OSHMPI can be utilized to leverage other runtimes (i.e., MPI) [out of scope]
5. Host-initiated operations on GPU memory



Design and Implementation



- Each processing element (PE) is mapped to a single PVC tile (for simplicity, PEs are mapped to whole device)
- Symmetric heap is allocated on each device tile
  - Allocated device heaps are registered to host back-end by extension APIs
  - Sandia OpenSHMEM (SOS) is used as the host back-end
- Intra-node inter-GPU transfers utilize the Xe Links for direct load / store
- Synchronization with host is achieved by a CPU proxy thread via a reverse offload ring buffer
  - 20 million requests per second with multiple GPU threads and a single CPU proxy
- Inter-node transfers use GPU RDMA using FI_HMEM capabilities of supported OFI providers

Device Extension APIs

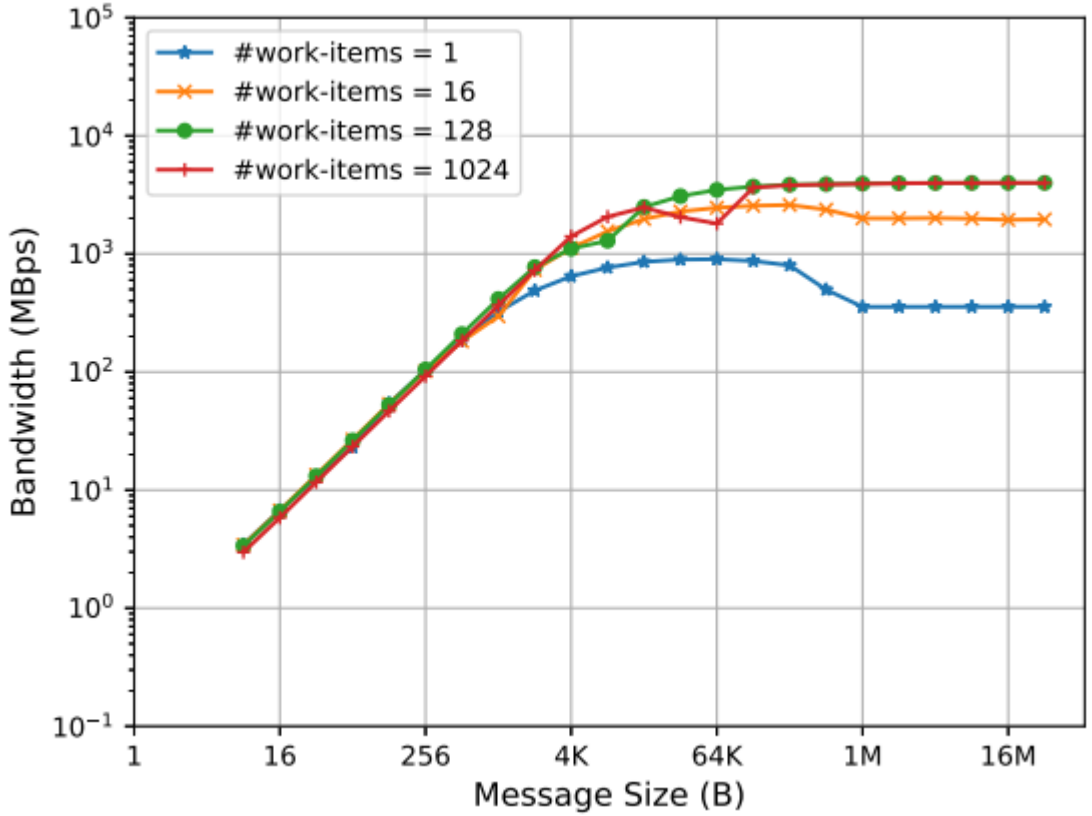
- GPUs launch group of threads to exploit the underlying SIMD/SIMT architecture
- We propose extensions that are callable from GPU kernel only

```
template<typename TYPE, typename Group>
void ishmemx_put_work_group(TYPE *dest, const TYPE *source, size_t nelems,
                            int pe, const Group &group)
```

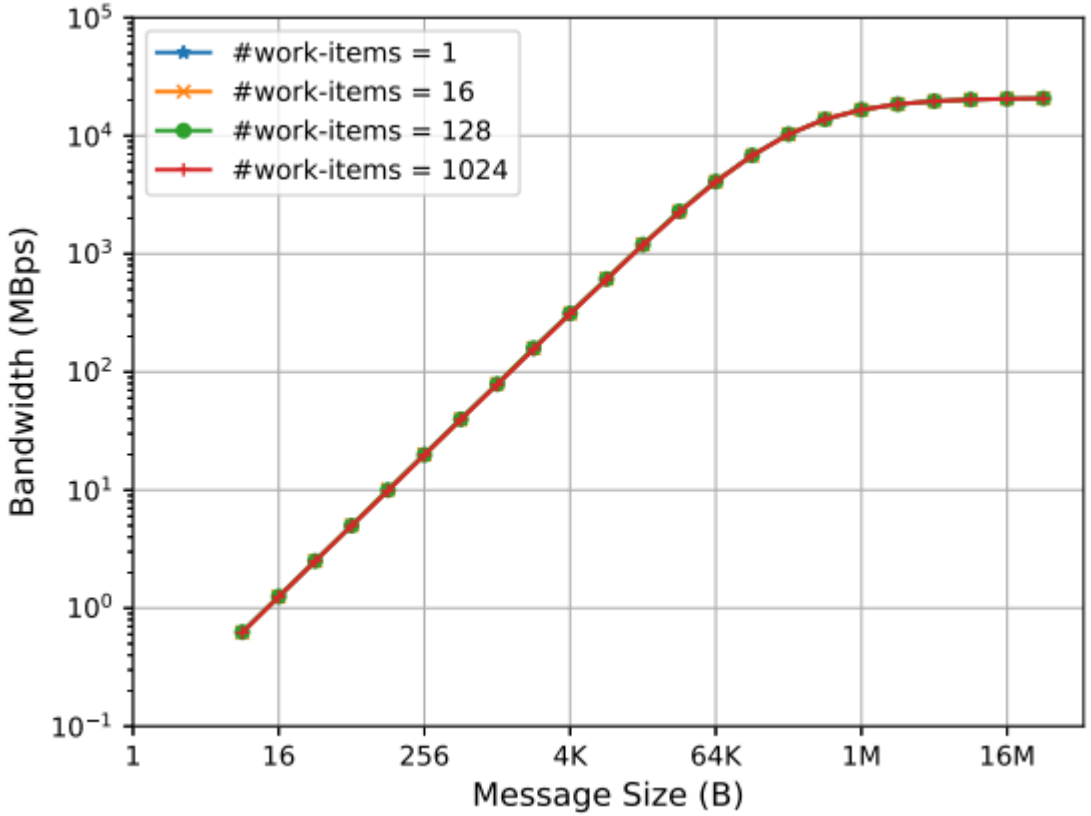
```
template<typename TYPE, typename Group>
int ishmemx_broadcast_work_group(ishmem_team_t team,
                                 TYPE *dest, const TYPE *source, size_t nelems,
                                 int pe, const Group &group)
```

- Requires a Group argument, which corresponds to SYCL work-group/sub-group
- Collective across work-items/threads in a Group
- For intra-node operations, work-items/threads in the Group can execute independently and in any order within the GPU, which optimizes performance
- For inter-node operations, a single designated thread initiate reverse offload, reducing contention

Store v/s Copy Engines (`ishmemx_put_work_group`)



Store from kernel work items



Copy engine transfer

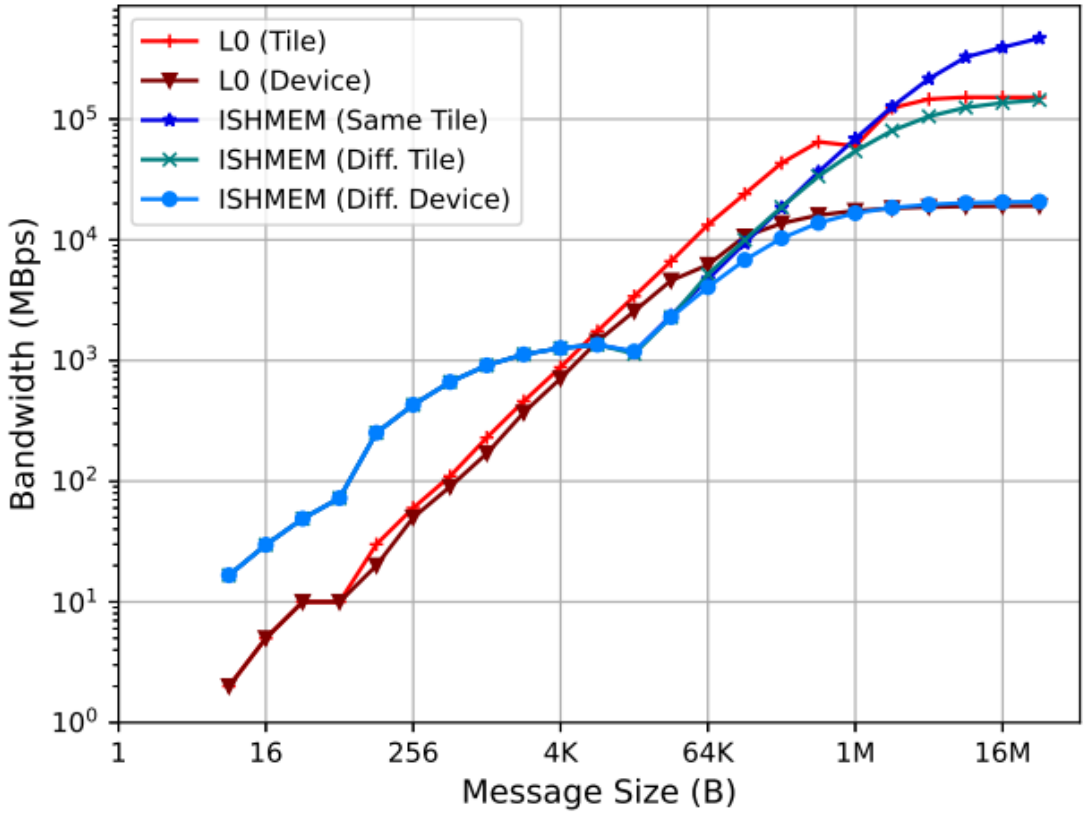
- Kernel driven direct load/store operation performs well for small to medium message sizes
- GPU copy engines provide higher bandwidth for larger message sizes
- Choosing appropriate cutover for performance depends on both message size and number of work-items

See slide #7 for workloads and configurations. Results may vary.

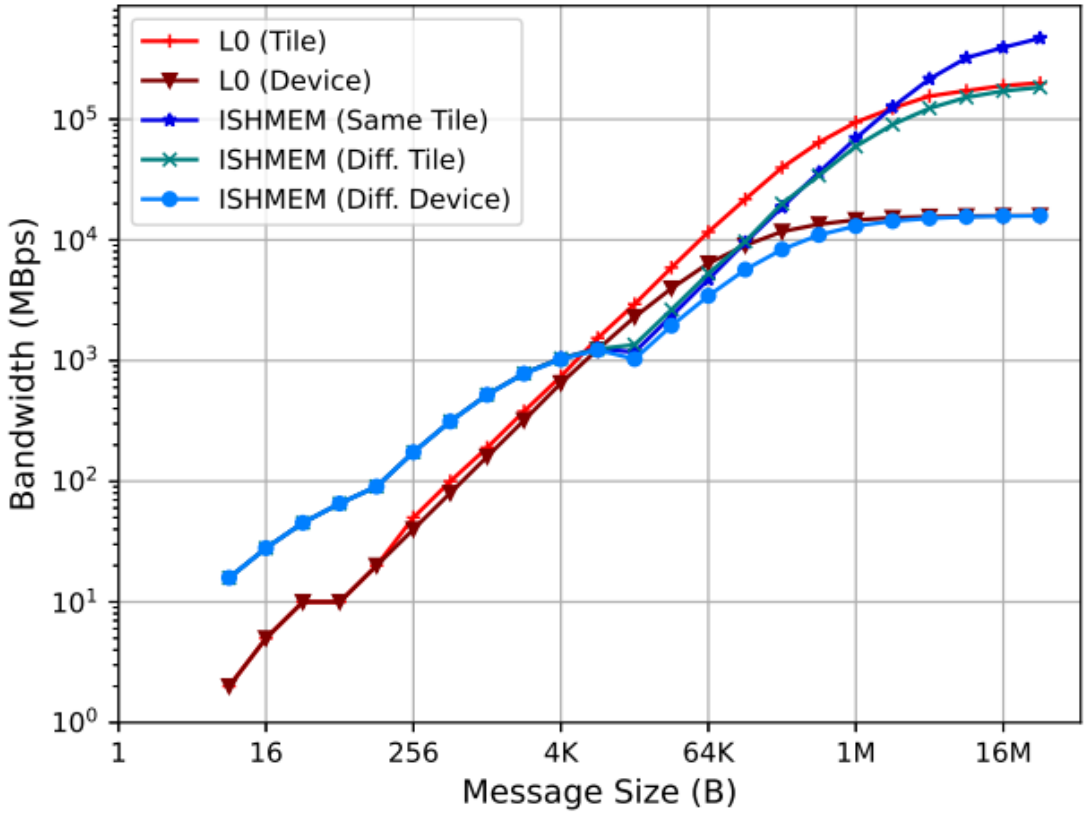
Experimental Setup

- Measurements done on Intel internal Aurora validation system, Borealis
- Hardware details:
  - 6 Intel® Data Center GPU Max Series (PVC) devices fully connected by Xe Link fabric
  - 2x Intel® Xeon® CPU Max 9470C (SPR) CPUs
    - 2.0 GHz, each socket with 52 cores and 2 threads per core
  - 8x HPE* Slingshot 11 NICs (200 Gbps each)
- OS: SUSE* Linux Enterprise Server 15 SP4
- Software Details:
  - Intel® SHMEM 1.1.0
  - Sandia OpenSHMEM (main branch)
  - Intel® oneAPI DPC++/C++ Compiler 2024.1.0
- Measurements are done using Intel® SHMEM performance benchmarks on August, 2024
- Performance results are replicated on Aurora Sunspot system

Performance – RMA (Standard API)



Device-initiated Put Bandwidth

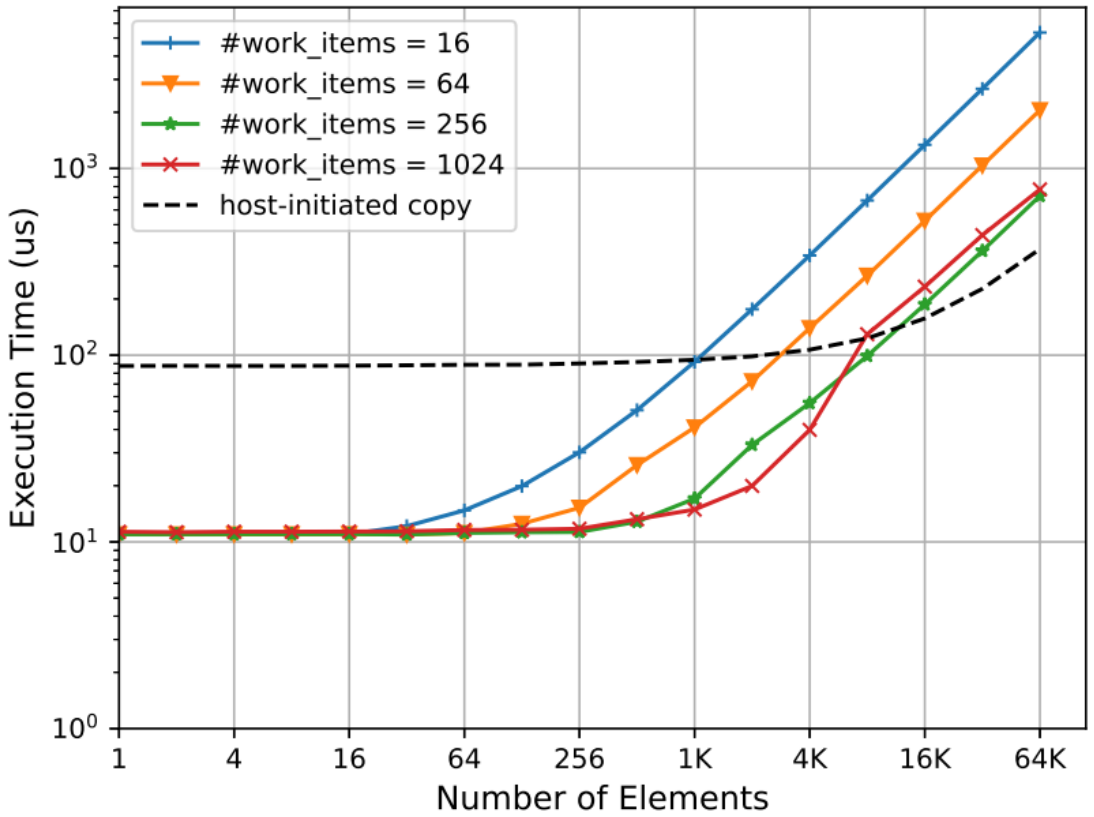


Device-initiated Get Bandwidth

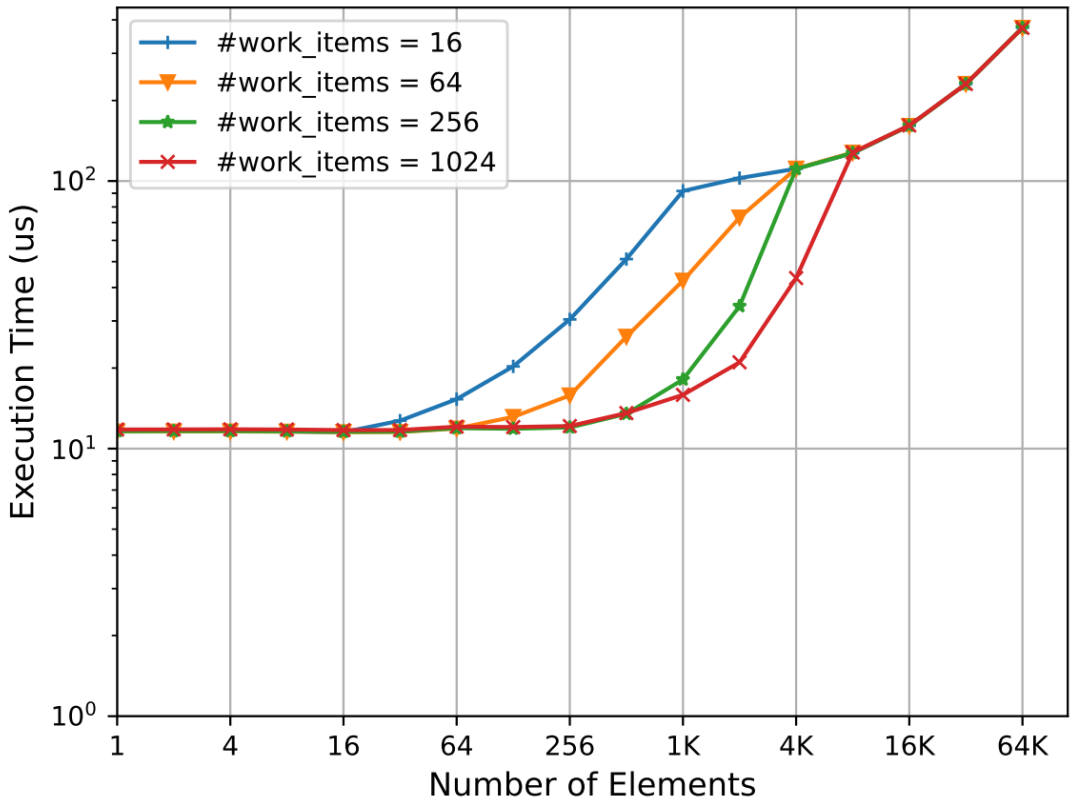
- Device-initiated operations perform better for up to 4KB message size; For larger message sizes, performance is similar to L0

See slide #7 for workloads and configurations. Results may vary.

Performance – Collective (Device Extension API)



Fcollect WG (12 PEs) – w/o cutover tuning



Fcollect WG (12 PEs) – w/ cutover tuning

- Number of work-items per work-group and total number of PEs affect the collective performance; Tuning for cutover ensures best performance across message sizes

See slide #7 for workloads and configurations. Results may vary.

Conclusion

- Intel® SHMEM 1.2.0 is available now
  - https://github.com/oneapi-src/ishmem
- GPU and Host initiated OpenSHMEM operations on GPU memory
- Optimized performance achieved by choosing appropriate cutover points between device driven load/store and host copy engine transfers
- Small operations are low latency; large operations achieve full hardware bandwidth

Intel® SHMEM: Scaling GPU Applications Up and Out

Dev Hub Tutorial at Intel Booth on Nov 20 10:30 AM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